

Natural numbers

The natural numbers are the counting numbers, starting at 1.

1, 2, 3, 4, ...

Is 0 a natural number under this convention?

Whole numbers

The whole numbers include 0 and the natural numbers.

0, 1, 2, 3, ...

Which whole number is not a natural number?

Integers

The integers include whole numbers and their negatives.

..., -3 , -2 , -1 , 0 , 1 , 2 , 3 , ...

Is every integer a whole number?

Rational numbers

A rational number is a ratio of integers with a nonzero denominator.

$$\frac{p}{q}, \quad p, q \in \mathbb{Z}, \quad q \neq 0$$

How could we write -6 as a ratio of integers?

Irrational numbers

An irrational real number cannot be written as a ratio of integers.

$$\sqrt{7}$$

What distinguishes its decimal expansion from a rational number's?

Real numbers

The real numbers include both rational and irrational numbers.

Where does the number 0 belong in the number sets we have discussed?

Classifying a number

$$-\frac{11}{4}$$

Which of our number sets contain this number?

Classifying a decimal

0.375

Is this number rational, and how could we justify our answer?

A radical and its value

$$\sqrt{81}$$

Which number sets contain the value of this expression?

Order of operations

$$18 - 2(3 + 1)^2$$

What value does this expression have?

Equal-precedence operations

$$24 \div 6 \cdot 2$$

What value results when we perform division and multiplication from left to right?

Grouping matters

$(-4)^2$ and -4^2

What value does each expression have?

Commutative properties

Commuting changes the order of addends or factors.

$$a + b = b + a \quad ab = ba$$

Does subtraction have the same property?

Associative properties

Associating changes grouping without changing order.

$$(a + b) + c = a + (b + c)$$

Does division have the same property?

Distributive property

Multiplication distributes over addition.

$$a(b + c) = ab + ac$$

How would we rewrite $5(x - 3)$ without parentheses?

Additive identity

An identity leaves a value unchanged under the operation.

$$a + 0 = a$$

What is the multiplicative identity?

Additive inverse

An additive inverse adds to its original value to give 0.

$$a + (-a) = 0$$

What is the additive inverse of -9?

Multiplicative inverse

A multiplicative inverse multiplies its original value to give 1.

$$a \cdot \frac{1}{a} = 1, \quad a \neq 0$$

Why must we exclude 0?

Evaluating an expression

Use $x = -2$.

$$3x^2 - 4x + 1$$

What is the value of the expression?

Like terms

Like terms have the same variable factors and exponents.

$$7x^2 - 3x + 2x^2 + 8x$$

What equivalent expression combines the like terms?

Simplifying an expression

$$4(2x - 3) - 3(x + 5)$$

What equivalent expression has no parentheses and no like terms left to combine?

Identifying a property

$$6(x + 2) = 6x + 12$$

Which property makes this equality true for every real x ?